

Question

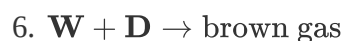
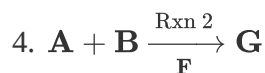
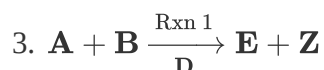
Salt **A** contains the metallic element **Z** with a mass fraction of 57.77%. When heated to 400°C , salt **A** decomposes with a weight loss of 11.144%, releasing the elemental gas **X**. The solid elemental substance **Y** can react with gas **X** in different proportions to form two binary compounds, **B** and **C**. **B** is a liquid with a mass ratio of 1.336 between its two elements, while **C** is a gas with a density approximately 9 times that of air under STP conditions.

A can dissolve in **B** to form a solution (denoted as Solution 1). When gas **D** is passed into Solution 1 (Reaction 1), **D** is oxidized to the cation D^{+} , and the elemental form of **Z** precipitates. The remaining part, after vacuum distillation, yields salt **E**, which shares the same cation D^{+} as **A**.

If **F** is added to Solution 1 instead, **A**, **B**, and **F** react in a 1:1:1 ratio to form the ionic compound **G** (Reaction 2). The structure of **G** is distinctive, as both its cation and anion belong to the O_h point group and are isoelectronic with each other.

Salt **H** has the same anion as **A**, but its molar mass is 2 g/mol greater than that of **A**. When **H** reacts with **B**, it releases the colorless gas **W** (Reaction 3). At room temperature, **W** can combine with **D** to form a reddish-brown gas, which can further dimerize.

The above transformations can be represented by the following reaction equations:



Based on the above information, select the most appropriate option from the following choices.

- A.** The elemental gas produced by the decomposition of **A** is O_2 , and the solid elemental **Y** belongs to a transition metal element.
- B.** The monatomic gas produced by the decomposition of **A** is O_2 , and the solid monatomic element **Y** belongs to the group IIIA elements.
- C.** The elemental gas produced by the decomposition of **A** is N_2 , and the solid elemental **Y** belongs to a transition metal element.
- D.** The elemental gas produced by the decomposition of **A** is N_2 , and the solid elemental **Y** belongs to the group IVA elements.
- E.** The elemental gas produced by the decomposition of **A** is Cl_2 , and the solid elemental **Y** belongs to the alkali metal or alkaline earth metal elements.
- F.** The monoatomic gas produced by the decomposition of **A** is Cl_2 , and the solid monoatomic **Y** belongs to a transition metal element.
- G.** The monatomic gas produced by the decomposition of **A** is Cl_2 , and the solid monatomic element **Y** belongs to the VA group elements.
- H.** The monoatomic gas produced by the decomposition of **A** is F_2 , and the solid monoatomic element **Y** belongs to the alkali metal or alkaline earth metal elements.
- I.** The elemental gas produced by the decomposition of **A** is F_2 , and the solid elemental **Y** belongs to the transition metal elements.

- J.** The elemental gas produced by the decomposition of **A** is F_2 , and the solid elemental **Y** belongs to the VIA group elements.
- K.** The monatomic gas produced by the decomposition of **A** is O_3 , and the solid monatomic **Y** belongs to an alkali metal or alkaline earth metal element.
- L.** The monoatomic gas produced by the decomposition of **A** is O_3 , and the solid monoatomic **Y** belongs to the Group VIIA elements.
- M.** The monatomic gas produced by the decomposition of **A** is a noble gas.
- N.** The cations in **G** are composed of Group VA and Group VIA elements, while the anions are composed of Group IVA and Group VIIA elements.
- O.** The cations in **G** are composed of Group VA and Group VIIA elements, while the anions are composed of Group VIA and Group VIIA elements.
- P.** The cations in **G** are composed of Group IVA and Group VIA elements, while the anions are composed of Group VA and Group VIA elements.
- Q.** In **G**, the cations are exclusively composed of Group IVA elements, while the anions are composed of Group VA and Group VIA elements.
- R.** In **G**, the cations are solely composed of Group VA elements, and the anions are composed of Group IVA and Group VIA elements.
- S.** The cations in **G** are exclusively composed of Group VIA elements, while the anions consist of Group VIA and Group VIIA elements.

- T.** The cations in **G** are composed solely of Group VIIA elements, while the anions are composed of Group VA and Group VIIA elements.
- U.** The cations in **G** are composed of Group VA and Group IVA elements, while the anions consist solely of Group IVA elements.
- V.** The cations in **G** are composed of Group VIA and Group VIIA elements, while the anions consist solely of Group VA elements.
- W.** The cations in **G** are composed of Group VIIA and Group VA elements, while the anions consist solely of Group VIA elements.
- X.** The cations in **G** are composed of Group VIIA and Group IIIA elements, while the anions consist solely of Group VIIA elements.
- Y.** None of the above answers are correct
- Z.** The above answers have two or more correct options

Answer

Correct Answer: **T**

Detailed Explanation

From the room-temperature combination of **W** and **D** yielding a red-brown dimeric gas, it can be inferred that dimerizable NO_2 gas should have been produced. From this and the fact that gas **D** can be oxidized to D^+ , it can be deduced that gas **D** should be NO , and thus **W** should be an oxidizing substance. It is easy to assume that **W** is O_2 ;

CHECKPOINT

0.5 PTS

Gas **D** should be NO

CHECKPOINT

0.5 PTS

W is O_2

B, C are two binary compounds, with the former having a mass ratio of 1.336; the latter has a density under standard atmospheric pressure that is 9 times that of air, which essentially means its molecular weight is approximately 9 times that of air's molecular weight (~ 29), i.e., the molecular weight of **C** is $261g/mol$.

CHECKPOINT

0.5 PTS

The molecular weight of **C** is 261g/mol

Gas **X** and solid elemental **Y** can combine to form **B, C**. Since **X** is not oxygen, the commonly considered elemental gases are F_2 or Cl_2 , as nitrogen gas rarely forms variable-valency binary compounds. A simple table calculation suggests that the corresponding elemental gas should be F_2 .

Number of F atoms x	Total relative atomic mass of F 19 x	Remaining mass 261 – 19	Possible combinations (other atom count y, other atom molecular mass M)	Matches elemental?	Mass ratio verification (xM : 19x' = 1.336)
1	19	242	(1, 242)	X	—
2	38	223	(1, 223)	X	—
3	57	204	(1, 204)	X	—
4	76	185	(1, 185)	X	—
5	95	166	(2, 83)	X	—
5	95	166	(1, 127)	X	—
6	114	147	(1, 147)	X	—
6	114	147	(3, 49)	X	—
7	133	128	(1, 128)	√ (I)	1\127 : 5\19 ≈ 1.336 (IF ₅)
8	152	109	(1, 109)	X	—
9	171	90	(1, 90)	X	—
10	190	71	(1, 71)	X	—

CHECKPOINT

1 PTS

The elemental gas **X** produced by the decomposition of **A** is F_2

From the mass ratio of **B**, it can be deduced that **B**, **C** should be IF_5 and IF_7 .

CHECKPOINT

0.5 PTS

B is IF_5 .

CHECKPOINT

0.5 PTS

C is IF_7 .

Based on the above information, the mass fraction of 57.77% can be used to calculate that the metal element **Z** is Au , and the salt **A** is $NO^+AuF_6^-$.

CHECKPOINT

1.0 PTS

A is $NO^+AuF_6^-$

Since salt **H** and salt **A** share the same anion, and **H** has a molar mass 2 g/mol greater than **A**, and **H** reacts with **B** to release colorless oxygen gas, it can be deduced that O_2^+ is the cation.

CHECKPOINT

1 PTS

O_2^+ is the cation of salt **H**

Salt **E** has the same cation as **A**, which is NO^+ , and its anion should originate from **B**. Thus, **E** is $NO^+IF_6^-$.

CHECKPOINT

1.0 PTS

Salt **E** is $NO^+IF_6^-$

Both the cation and anion of **G** belong to the O_h point group and are isoelectronic. Since both the cation and anion belong to the O_h point group, they should both be of the form $XF_6^\pm$ (where X represents some element).

Based on the reaction equation for the formation of **G**, **G** must contain either IF_6^- or IF_6^+ . If it were IF_6^- , there is no reasonable isoelectronic cation, so it can be inferred that the cation of **G** is IF_6^+ .

CHECKPOINT

1 PTS

The cation of **G** is IF_6^+

The anion is an isoelectronic counterpart of IF_6^+ . Since the reaction equation is 1:1:1, **G** must be a 1:1 salt, meaning the anion can only be SbF_6^- . Thus, **G** is $IF_6^+SbF_6^-$.

CHECKPOINT

0.5 PTS

G must be a 1:1 salt

CHECKPOINT

1.0 PTS

G is $IF_6^+ SbF_6^-$.

Based on the material inference results, only option T is correct.